Highlights

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales

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- Mammalian abdominal resonance governed by quasi-universal $\Pi_0 \approx 0.07$
- 1458-point parametric study confirms algebraic consistency of non-dimensionalisation
- Cross-species f_2 scales as $(1/a)\sqrt{E/\rho_f}$ when other Π-groups are approximately constant
- Scattering coupling ratio $\mathcal{R}_{\text{scat}} \sim 10^3$–$10^4$ is size-independent: animal models exhibit similar scaling under mechanical excitation
- Breathing mode never reaches infrasound for any organism

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales

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Abstract

Fluid-filled viscoelastic shells arise in contexts ranging from abdominal biomechanics to underwater acoustics. Modal frequencies of such shells depend on ten physical parameters, making systematic comparison across geometries and species impractical without dimensional reduction. We apply Buckingham Pi analysis to the governing equations of a fluid-filled oblate spheroidal shell, identifying eight dimensionless groups of which five govern the undamped flexural mode frequencies. The resulting scaling law, $f_n = \sqrt{E/\rho_f}\,(1/a)\,\Phi_n(\Pi_1, \Pi_2, \Pi_3, \Pi_5, \Pi_6)$, where Φ_n absorbs the $1/(2\pi)$ factor, admits a closed-form expression from the shell equations. A 1458-point parametric study spanning physiological ranges of all five governing dimensionless groups collapses onto the analytical curves to within numerical round-off (7.9×10^{-16}), confirming the algebraic consistency of the non-dimensionalisation. Cross-species predictions for four mammals (rat, cat, pig, human) reveal a quasi-universal dimensionless frequency $\Pi_0 \approx 0.07$, implying that the scattering coupling ratio $\mathcal{R}_{\text{scat}} = 1/(kR_{\text{eq}})^2 \sim 10^3$–$10^4$ is approximately size-independent and that animal models exhibit similar scaling behaviour to the human abdomen, provided excitation is delivered mechanically rather than acoustically. The breathing mode ($n = 0$), dominated by the fluid bulk modulus, would require a body radius of order 20 m to reach 20 Hz and is therefore irrelevant to infrasound for any biological organism.

Keywords: dimensional analysis, fluid-filled shell, scaling laws, viscoelastic, abdominal resonance, Buckingham Pi

1. Introduction

The modal frequencies of fluid-filled viscoelastic shells depend on at least ten physical parameters: geometric (semi-axes a, c; wall thickness h), material (elastic modulus E, Poisson's ratio ν, wall density ρ_w, loss tangent η), fluid (ρ_f, bulk modulus K_f), and loading (intra-abdominal pressure P_{iap}) [1, 2, 3, 4, 5, 6]. This high-dimensional parameter space makes comparison across species, organs, and loading conditions impractical without systematic dimensional reduction.

In a companion paper [7], we developed a fluid-filled oblate spheroidal shell model for the human abdominal cavity, predicting flexural resonances at 4 Hz to 10 Hz and a mechanical-to-airborne full displacement ratio $\mathcal{R}_{\text{full}} \approx 6.6 \times 10^4$. That metric combines the Rayleigh scattering penalty with the air–tissue impedance mismatch, whereas the present paper isolates only the scattering contribution $\mathcal{R}_{\text{scat}}$. Here we show that mammalian abdominal resonance is governed by a quasi-universal dimensionless frequency $\Pi_0 \approx 0.07$, implying that animal models exhibit similar scaling behaviour and that the scattering coupling ratio $\mathcal{R}_{\text{scat}}$ is approximately size-independent. These results follow from a

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Buckingham Pi reduction of the ten-parameter model (plus the output frequency, giving eleven variables) to five governing dimensionless groups for flexural modes.

Three questions motivate this analysis. First, can animal models (rat, cat, pig) be used to study abdominal resonance phenomena that manifest in humans? The answer depends on whether the governing dimensionless groups—and hence $\mathcal{R}_{\text{scat}}$—are approximately conserved across species. Second, can the breathing mode ($n = 0$)—whose frequency is dominated by the fluid bulk modulus—ever enter the infrasonic range for any biological organism? Third, which dimensionless groups dominate the flexural resonance, and how sensitive is f_2 to each?

2. Dimensional analysis

2.1. Governing variables

The natural frequency f_n of mode n and its ten governing parameters form a set of eleven variables: $f_n = f(E, a, c, h, \nu, \rho_w, \rho_f, K_f, P_{\text{iap}}, \eta)$. With three fundamental dimensions (M, L, T), the Buckingham Pi theorem [8] yields $11 - 3 = 8$ independent dimensionless groups (Table 1).

Table 1: Dimensionless groups from Buckingham Pi theorem. Groups marked $*$ drop out of the flexural-mode frequency.

Group	Definition	Name	Role
Π_0	$f_n\, a\sqrt{\rho_f/E}$	dimensionless frequency	output
Π_1	h/a	thickness ratio	stiffness & mass
Π_2	c/a	aspect ratio	geometry
Π_3	ρ_w/ρ_f	density ratio	added mass
Π_4^*	K_f/E	compressibility ratio	breathing mode only
Π_5	P_{iap}/E	prestress ratio	tension stiffening
Π_6	ν	Poisson's ratio	membrane coupling
Π_7^*	η	loss tangent	damping only

2.2. Reduction for flexural modes

For flexural modes ($n \geq 2$), two simplifications apply. The fluid bulk modulus K_f contributes no restoring force (the fluid acts as added mass only), so Π_4 drops out. The loss tangent η affects the quality factor but not the undamped frequency, so Π_7 drops out. The effective parameter count reduces to six: a dimensionless frequency output Π_0 governed by five input groups ($\Pi_1, \Pi_2, \Pi_3, \Pi_5, \Pi_6$).

The scaling law takes the closed form

$$f_n = \sqrt{\frac{E}{\rho_f}}\,\frac{1}{a}\,\Phi_n\left(\frac{h}{a},\ \frac{c}{a},\ \frac{\rho_w}{\rho_f},\ \frac{P_{\text{iap}}}{E},\ \nu\right), \tag{1}$$

where the $1/(2\pi)$ factor is absorbed into Φ_n, which admits a closed-form expression from the equivalent-sphere shell equations:

$$\Phi_n = \frac{1}{2\pi}\sqrt{\frac{K_{\text{bend}} + K_{\text{memb}} + K_{\text{pre}}}{m_{\text{eff}}}} \cdot a\sqrt{\frac{\rho_f}{E}}, \tag{2}$$

with

$$K_{\text{bend}} = n(n-1)(n+2)^2\,\frac{D}{R^4}, \tag{3}$$

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \quad K_{\text{pre}} = \frac{P_{\text{iap}}}{R}(n-1)(n+2), \tag{4}$$

$$m_{\text{eff}} = \rho_w h + \frac{\rho_f R}{n}, \tag{5}$$

where $D = Eh^3/[12(1-\nu^2)]$ is the flexural rigidity, $R = (a^2c)^{1/3}$ is the equivalent-sphere radius, and the membrane mode-shape factor is

$$\lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{6}$$

2.3. Sensitivity: which Π-group dominates?

For the $n = 2$ mode at canonical parameters ($h/a = 0.056$, $c/a = 0.67$, $\nu = 0.45$, $P_{\text{iap}}/E = 0.01$), the stiffness budget is shared between the membrane term K_{memb} (~54%) and the prestress term K_{pre} (~45%), with bending contributing less than 2%. The transition to bending-dominated behaviour occurs near $h/a \approx 0.42$, well above the physiological range.

Consequently, f_2 is most sensitive to the thickness ratio $\Pi_1 = h/a$, which enters both the membrane stiffness (linearly through h) and the effective mass (linearly through $\rho_w h$). The prestress ratio $\Pi_5 = P_{\text{iap}}/E$ contributes substantially at canonical values but is approximately constant across mammals ($P/E \approx 0.006$–0.01), so it does not drive inter-species variation. The density ratio $\Pi_3 = \rho_w/\rho_f$ has a more modest effect: because soft tissue and peritoneal fluid have similar densities ($\Pi_3 \approx 1.04$–1.08), the fluid added mass $\rho_f R/n$ dominates the denominator of m_{eff}. This sensitivity structure explains why Π_0 is quasi-universal across mammals: the governing ratios h/a, ρ_w/ρ_f, and P/E are constrained by shared developmental biology [9].

3. Parametric validation

We evaluate (1) over a 1458-point grid spanning physiological ranges of all five governing dimensionless groups: $h/a \in [0.025, 0.100]$, $c/a \in [0.50, 0.80]$, $\rho_w/\rho_f \in [0.98, 1.18]$, $P_{\text{iap}}/E \in [5 \times 10^{-4}, 0.02]$, and $\nu \in [0.40, 0.49]$. In dimensional terms these correspond to $E \in [0.05, 2.0]$ MPa (6 values), $a \in [0.15, 0.20]$ m (3 values), $c/a \in [0.5, 0.8]$ (3 values), $h \in [0.005, 0.015]$ m (3 values), $\rho_w/\rho_f \in [0.98, 1.18]$ (3 values), and $\nu \in [0.40, 0.49]$ (3 values), with K_f and P_{iap} held at canonical values. The loss tangent η is excluded because it does not affect the undamped natural frequency.

Figure 1 shows the result. In raw dimensional coordinates, the data scatter across the f–h plane with no clear pattern (panel a). Non-dimensionalisation via (1) collapses all 1458 points onto universal curves $\Pi_0(h/a)$ parameterised by c/a (panel b). The parity plot (panel c) confirms that the analytical Φ_n reproduces every numerical

point to within numerical round-off (maximum relative error 7.9×10^{-16}), confirming the algebraic consistency of the non-dimensionalisation rather than providing independent physical validation, which was established in the companion paper [7].

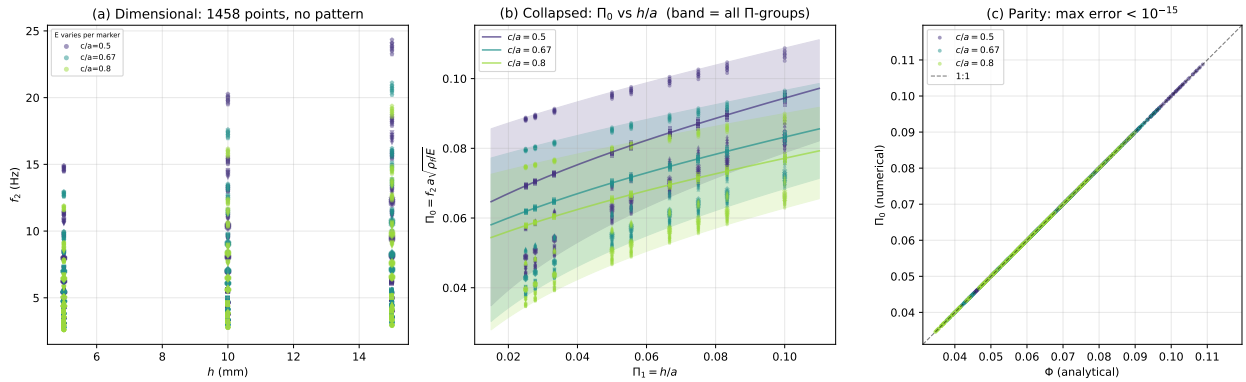

Figure 1: Dimensional collapse of the 1458-point parametric study. *(a)* Raw dimensional data (f_2 vs. h) shows no clear pattern. *(b)* Non-dimensionalising via equation (1) collapses all points onto universal curves $\Pi_0(h/a)$ parameterised by c/a; shaded bands span the ranges of P_{iap}/E, ρ_w/ρ_f, and ν. *(c)* Parity plot confirms the analytical Φ reproduces every numerical point to within numerical round-off, confirming algebraic consistency.

4. Cross-species scaling

The scaling law (1) predicts modal frequencies for any organism given its body dimensions and tissue properties. Table 2 lists predictions for four mammals, using species-specific parameters estimated from broad allometric trends and tissue-mechanics ranges rather than direct organ-level measurements of the abdominal wall in each species [10, 11, 9]. These values should therefore be read as scaling estimates rather than as definitive anatomical measurements. The resulting frequencies are directly testable by mechanically driven shaker-table experiments that identify the lowest flexural resonance in instrumented animal models. Four of the five governing dimensionless groups (h/a, c/a, ρ_w/ρ_f, P_{iap}/E) are shown explicitly; Poisson's ratio ($\nu = 0.45$) is held constant. The quantity $kR_{\text{eq}} = 2\pi f_2 R_{\text{eq}}/c_{\text{air}}$, where $R_{\text{eq}} = (a^2 c)^{1/3}$ is the equivalent-sphere radius, is the Helmholtz number governing airborne coupling.

Three observations emerge. First, the dimensionless frequency Π_0 is approximately constant (0.065–0.072) across a six-fold range of body size, confirming that the governing Π-groups take similar values across mammalian species. The coefficient of variation across four species is 4.5%, smaller than the approximately 20% uncertainty in any individual E or h measurement. Second, the scattering coupling ratio $\mathcal{R}_{\text{scat}}$ varies by only a factor of two (from 7.7×10^3 for humans to 1.7×10^4 for rats), remaining firmly in the 10^3–10^4 range. For the present $n = 2$ flexural mode, $\mathcal{R}_{\text{scat}}$ is the Rayleigh scattering component of the fuller ratio $\mathcal{R}_{\text{full}}$ reported in Paper 1 [7]; higher-order modes inherit a stronger $(kR_{\text{eq}})^n$ penalty. Airborne coupling is negligible for all species studied. Third, the product kR_{eq} is of order 10^{-2} for all four species, placing every organism deep in the Rayleigh regime.

Table 2: Cross-species scaling of the $n = 2$ flexural mode. Four of the five governing Π-groups are shown; $\nu = 0.45$ is held constant across species. $\mathcal{R}_{\text{scat}} = 1/(kR_{\text{eq}})^2$ quantifies the Rayleigh-regime scattering penalty for airborne excitation; for the present $n = 2$ mode it is the geometric Rayleigh factor within the fuller airborne-to-mechanical ratio reported in [7], whose prefactor and $(kR_{\text{eq}})^n$ dependence vary with modal order. The full displacement ratio in [7] is approximately 8× larger because it additionally includes the acoustic impedance mismatch between air and tissue. Allometric estimates of E, h/a, and c/a are inferred from broad tissue-mechanics and size-scaling literature [10, 11]; uncertainties in any individual E or h estimate propagate to approximately 20% uncertainty in f_2 via the $\sqrt{E}$ dependence.

Species	a (cm)	E (MPa)	h/a	c/a	ρ_w/ρ_f	P_{iap}/E	f_2 (Hz)	Π_0	kR_{eq}	$\mathcal{R}_{\text{scat}}$
Rat	3	0.05	0.067	0.70	1.04	0.006	15.7	0.067	7.7×10^{-3}	1.7×10^4
Cat	8	0.08	0.050	0.70	1.05	0.008	7.2	0.065	9.3×10^{-3}	1.2×10^4
Pig	15	0.10	0.053	0.65	1.06	0.008	4.5	0.069	1.1×10^{-2}	8.6×10^3
Human	18	0.10	0.056	0.67	1.08	0.010	3.9	0.072	1.1×10^{-2}	7.7×10^3

This size-independence of $\mathcal{R}_{\text{scat}}$ has a practical implication: animal models exhibit similar scaling behaviour to the human abdomen in studies [12, 13], *provided excitation is delivered mechanically* (e.g. shaker table) rather than acoustically. Acoustic experiments would require compensating for the $(kR_{\text{eq}})^n$ coupling penalty, which varies with body size and frequency.

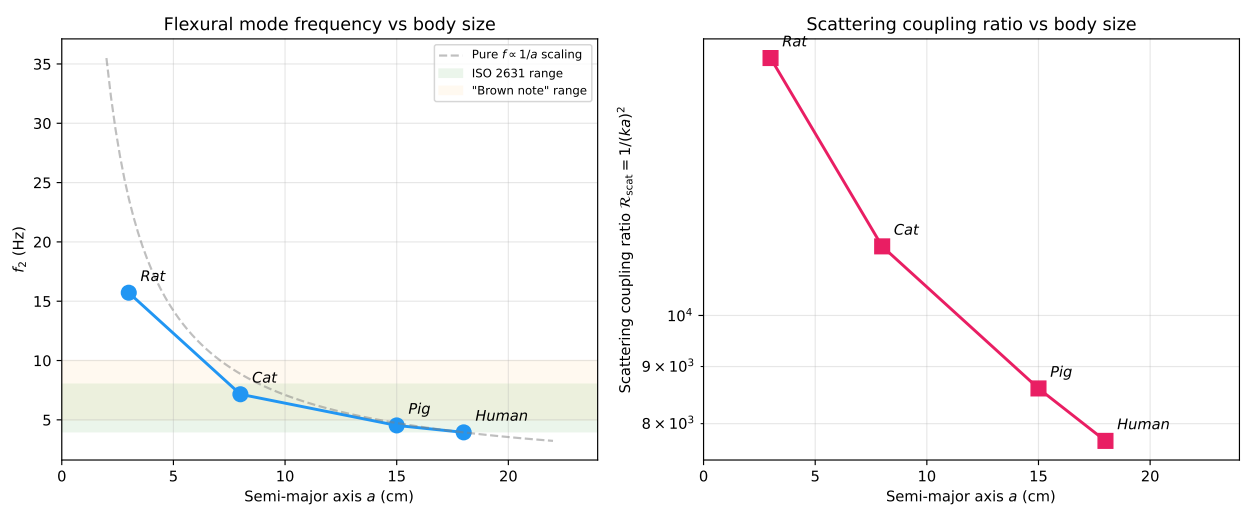

Figure 2: Cross-species scaling. *Left:* Flexural mode f_2 vs. semi-major axis a. The dashed line shows pure $1/a$ scaling; deviations arise because smaller animals have softer tissue and different h/a. *Right:* Scattering coupling ratio $\mathcal{R}_{\text{scat}} = 1/(kR_{\text{eq}})^2$ is approximately size-independent, supporting the validity of animal-model extrapolation.

5. The breathing mode is never infrasonic

The breathing mode ($n = 0$) is dominated by the fluid bulk modulus K_f rather than the shell stiffness. Its frequency is approximately

$$f_0 \approx \frac{1}{2\pi}\sqrt{\frac{3K_f}{\rho_f R^2}}, \tag{7}$$

where K_f is the fluid bulk modulus, ρ_f is the fluid density, and R is the equivalent radius. In the fluid-dominated limit, this gives the same scaling $f_0 \propto \sqrt{K_f/\rho_f}/R$; evaluated at the canonical human R_{eq} it yields $f_0 \approx 2.57\,\mathrm{kHz}$, while the full fluid–shell model gives $f_0 \approx 2490\,\mathrm{Hz}$. To bring f_0 down to 20 Hz—the upper boundary of infrasound—would require an equivalent radius of order 20 m. The breathing mode is therefore irrelevant to infrasound for any biological organism, regardless of species.

6. Concluding remarks

Buckingham Pi analysis reduces the ten-parameter fluid-filled viscoelastic shell model (eleven variables including the output frequency) to five governing dimensionless groups for flexural modes. The resulting closed-form scaling law collapses a 1458-point parametric study with algebraic consistency and provides cross-species predictions without re-solving the eigenvalue problem. The dimensionless frequency $\Pi_0 \approx 0.07$ is quasi-universal across mammals, and the scattering coupling ratio $\mathcal{R}_{\mathrm{scat}} \sim 10^3$–$10^4$ is approximately size-independent, supporting the use of animal-model scaling for abdominal resonance studies under mechanical excitation. The breathing mode never enters the infrasonic range for any organism.

The dimensionless framework developed here has potential applications beyond abdominal biomechanics. In blast-injury research, the scaling law may enable direct extrapolation from animal-model data to human injury thresholds without repeating experiments at human scale. In medical ultrasound, fluid-filled organ models are routinely used to predict diagnostic and therapeutic responses; the Π-group reduction offers a systematic basis for cross-organ and cross-species comparison [14, 15]. In vehicle NVH assessment, whole-body vibration standards [13] rely on empirical transfer functions whose mechanistic basis could benefit from scaling laws of the type derived here. In underwater acoustics, fluid-filled elastic shells are canonical targets for sonar modelling [4, 3]; the present dimensional reduction applies directly to that class of problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The analysis code and parametric data are available at `https://github.com/JonathanMace/brownnote`.

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